

YEAR 7 MATHEMATICS · CHAPTER 10

Analysing dot plots and stem-and-leaf plots

Lesson 7 of 8



In this lesson you will learn to:

- find the range, median, mode and mean from a dot plot
- find the same statistics from a stem-and-leaf plot
- choose the best measure for data with an outlier

Name

Class

Teacher

Before you start

You will need

Pencil

Ruler

Calculator

Syllabus: MA4-DAT-C-02: analyses simple datasets using measures of centre, range and shape of the data

WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1** Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2** At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

Lesson map

Page	What is on it	Done
3	Learning intentions, key terms, notes and Level 1 teacher examples	☐
4	Level 1: YOU DO practice questions	☐
5	Level 2: WE DO teacher examples	☐
6	Level 2: YOU DO practice questions	☐
7	Level 3: WE DO teacher examples and YOU DO practice questions	☐
8	Level 3: YOU DO practice questions	☐
9	Extension questions and the tear-off exit ticket	☐
10	Lesson summary: a worked example for every level	☐

YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

- 1** Mean of 2, 4, 6

- 3** $0 \times 3 + 1 \times 5 + 2 \times 4$

- 2** Median of 3, 1, 8, 5

- 4** Range of 12, 30, 21



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON
10.07

Analysing dot plots and stem-and-leaf plots

Date _____

Learning intentions

We are learning to:

- find the range, median, mode and mean from a dot plot
- find the same statistics from a stem-and-leaf plot
- choose the best measure for data with an outlier

Success criteria

I can:

- ☐ use the **frequency** of each value to find the total and the mean
- ☐ count to the middle position of a plot to find the **median**
- ☐ explain how an **outlier** changes the mean and median

Key terms Write the definition as your teacher explains it.

Frequency

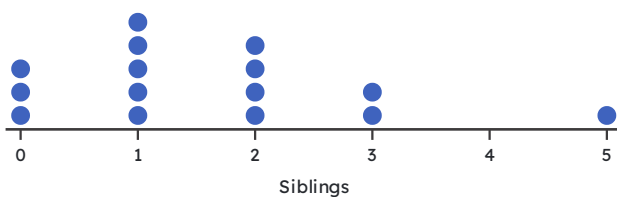
Measure of centre

Spread

Notes

Level 1 · Getting started

Number of siblings



WE DO

With your teacher: copy what they write on the board.

Example 1

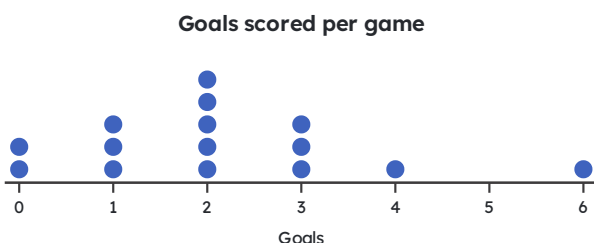
Find the range and the mode.

Example 2

Find the median and the mean.

YOU DO

On your own: Set A. Use the goals dot plot.



- 1 How many games were played?
- 2 What is the most goals in a game?
- 3 What is the range?
- 4 What is the mode?
- 5 What is the median?
- 6 How many goals were scored in total?
- 7 What is the mean (2 decimal places)?
- 8 Which value is an outlier?
- 9 In how many games were more than 2 goals scored?
- 10 What fraction of games had 0 goals?

YOU DO

On your own: Set B. Ages of people in a yoga class.

Stem	Leaf
2	4 6
3	1 1 5 8
4	0 2 7
5	3

Key: 3 | 1 = 31 years

- 1 How many people are in the class?
- 2 Who is the youngest?
- 3 Who is the oldest?
- 4 What is the range?
- 5 What is the mode?
- 6 What is the median?
- 7 What is the total of the ages?
- 8 What is the mean age?
- 9 How many are under 30?
- 10 What does 3 | 5 stand for?



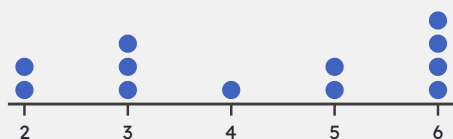
Level 2 · Building up

WE DO

With your teacher: copy what they write on the board.

Example 3

Find the range, median, mode and mean.

**Example 4**

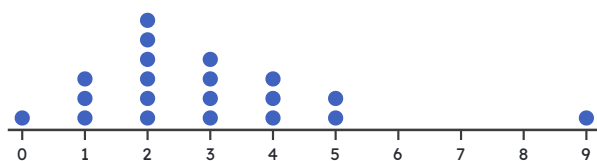
Find the range, median, mode and mean.

Stem	Leaf
1	5 8
2	0 3 3 7
3	1 4 4 4 9
4	2

Key: 2 | 3 = 23

YOU DO

On your own: Set C. Questions 1–5 use the books dot plot and 6–10 use the homework plot.

Books read in a month**1** Books: find the range.**2** Books: find the mode.**3** Books: find the median.**4** Books: find the mean.**5** Books: which value is an outlier? Does it make the mean bigger or smaller?**Minutes spent on homework**

Stem	Leaf
1	0 5 5 8
2	0 2 5 5 5 9
3	0 4 5
4	5

Key: 2 | 5 = 25 min

6 Homework: find the range.**7** Homework: find the mode.**8** Homework: find the median.**9** Homework: find the mean (2 d.p.).**10** Homework: how many students spent more than 30 minutes?



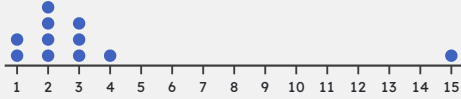
Level 3 · Challenge

WE DO

With your teacher: copy what they write on the board.

Example 5

Find the mean and median. Then remove the outlier and find them again.

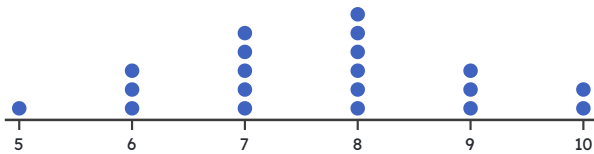
**Example 6**

Which measure (mean, median or mode) best describes the data in Example 5? Explain.

YOU DO

On your own: Set D. Questions 1–4 use the test scores dot plot.

Test scores out of 10



- 1 Find the mean, median, mode and range of the test scores.

- 2 A student who was away scores 2. Find the new mean and median.

- 3 Which measure changed more in Question 2? Why?

- 4 Could one more student raise the class mean to 8? Explain.

YOU DO

On your own: Set D continued. Questions 5–8 use the concert plot.

Ages at a concert	
Stem	Leaf
1	5 6 7 7 8 9
2	0 1 1 1 3 5
3	2 4
4	
5	
6	8

Key: 2 | 1 = 21 years

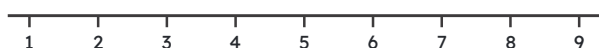
- 5 Find the mean, median, mode and range of the concert ages.

- 6 Which age is an outlier? How can you tell?

- 8 Which measure best describes a typical age at the concert? Why?

- 7 Remove the outlier. Find the new mean and range.

- 9 Draw a dot plot of 9 values with mode 4, median 5, range 6 and smallest value 2.



- 10 Why are the mean and median of a symmetric dot plot about the same?

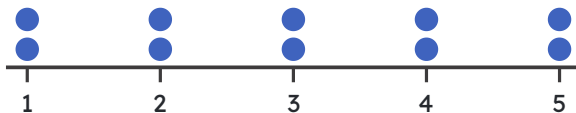
Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

A dot plot shows 10 values with a mean of 3. Nine of the values are 1, 2, 2, 3, 3, 4, 4, 5, 5. What is the tenth value?

1. Total of all 10 values: $10 \times 3 = 30$
2. Total of the nine values: 29
3. Tenth value: $30 - 29 = 1$



Answer: 1

YOU DO

On your own: read the worked example, then try these.

- E1** A stem-and-leaf plot shows 23, 25, 2?, 30, 34. The mean is 28. What is the missing leaf?

- E2** Add one value to 4, 5, 5, 6, 9 so that the median becomes 5.5. Which values work?

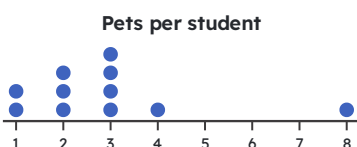
- E3** Make up six values where the mean is bigger than the median. Explain how you did it.

✕ Show off your skill: tear along this line and hand it to your teacher ✕

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

What is the mode?

Level 2

Find the median and the range.

Level 3

Find the mean. What happens to the mean if the 8 is removed?

Lesson summary

Missed the lesson? Start here.

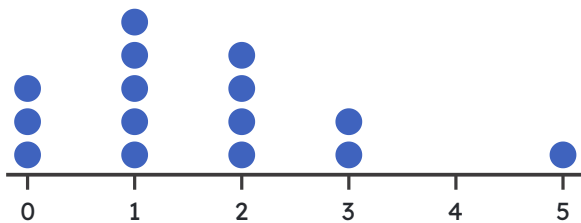
- From a plot**
1. **Range:** highest – lowest.
 3. **Median:** count to the middle position.

2. **Mode:** tallest stack or most repeated leaf.
4. **Mean:** add every value (value × frequency), then divide.

Level 1

Find the range and mode of the siblings dot plot.

1. Range: highest – lowest = $5 - 0 = 5$
2. The tallest stack is above 1



Answer: range 5, mode 1

Level 2

Find the median of this plot.

1. There are 7 values, so the median is the 4th
2. Count along: 21, 24, 24, 30

Stem	Leaf
2	1 4 4
3	0 5
4	2 6

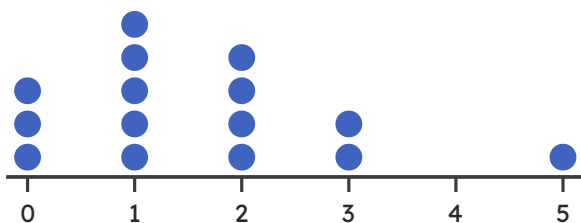
Key: 2 | 1 = 21

Answer: 30

Level 3

Find the mean of the siblings dot plot.

1. Total: $0 \times 3 + 1 \times 5 + 2 \times 4 + 3 \times 2 + 5 \times 1 = 24$
2. There are 15 values: $24 \div 15 = 1.6$

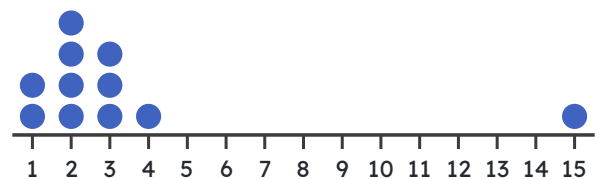


Answer: 1.6

Extension

Why does an outlier change the mean more than the median?

1. The mean adds up every value, so one big value adds a lot
2. The median only uses the middle position



Answer: the outlier pulls the mean towards it

This space is left blank on purpose. It is the back of your exit ticket.